http://home.pf.jcu.cz/~kyklop/SERYM/automatizace/jer/Kap06/Kap_06.htm

1 Obecně

Přenos regulátoru $R(p) = \frac{X(p)}{E(p)}$

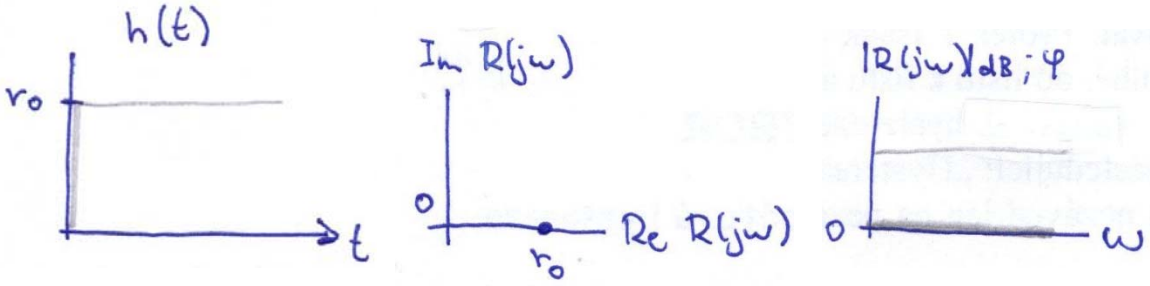
Gaussova rovina vyjadřuje frekvenční přenos R(jω).

Frekvenční charakteristiky zobrazují $|R(j\omega)|_{dB}$ a fázový posuv.

2 Proporcionální

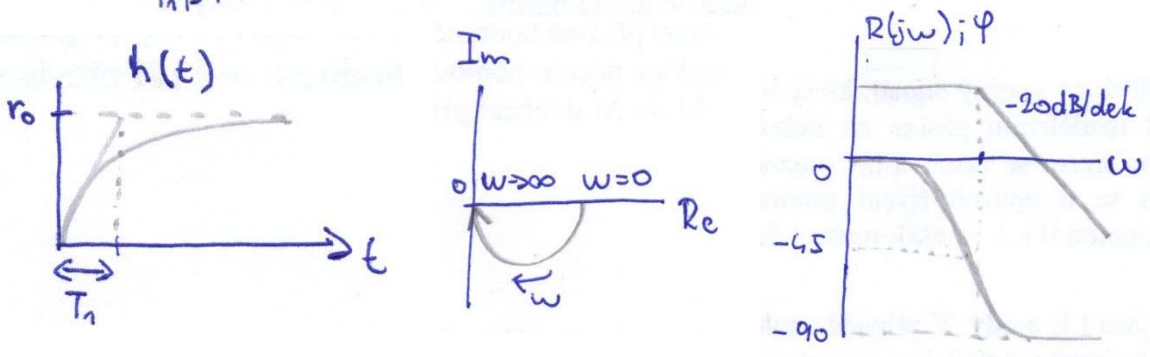
2.1 Bez zpoždění – 0. řád

$R(p) = r_0$ $\quad x(t) = r_0 \cdot e(t)$



2.2 Se zpožděním 1. řád

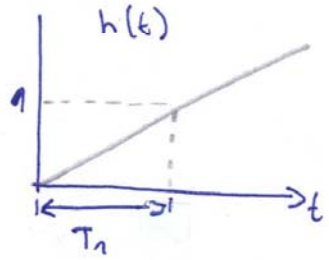
$R(p) = \frac{r_0}{T_1 p + 1}$ $\quad T_1 \cdot x'(t) + x(t) = r_0 \cdot e(t)$

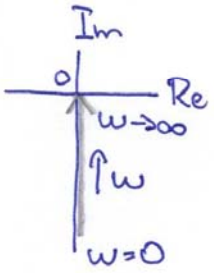


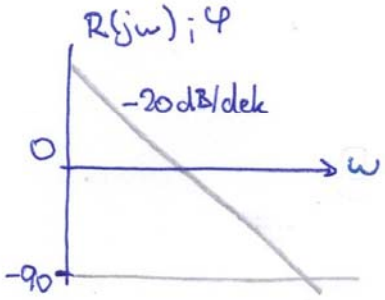
3 Integrační

3.1 Bez zpoždění – 0.řád

$$R(p) = \frac{1}{T_1 p} \qquad x(t) = \frac{1}{T_1} \cdot \int_0^t e(\tau)\, d\tau$$

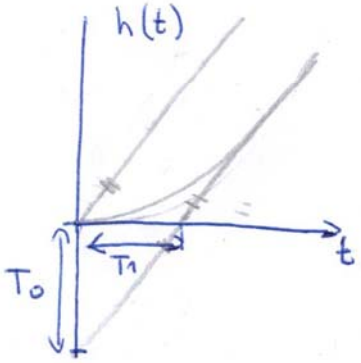


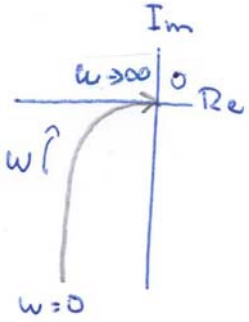


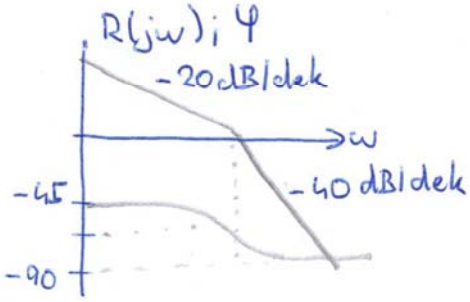


3.2 Se zpožděním 1. řád

$$R(p) = \frac{1}{p(T_1 p + T_0)} \qquad T_1 x'(t) + T_0 x(t) = \int_0^t e(\tau)\, d\tau$$



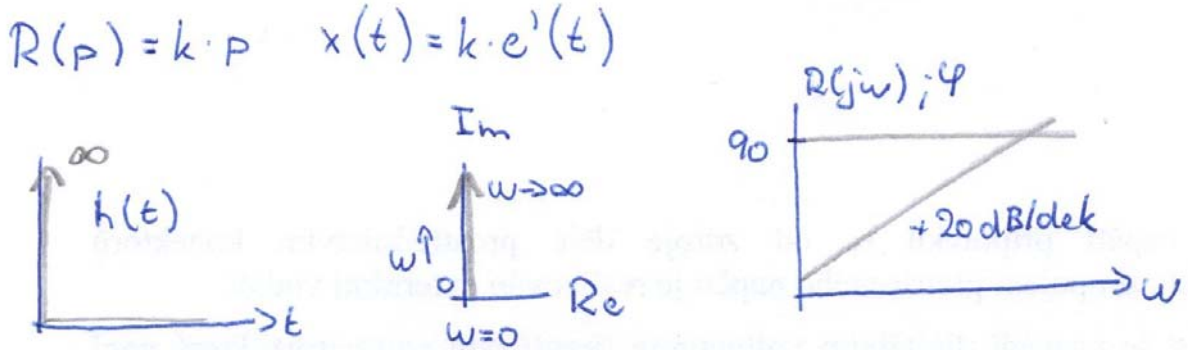




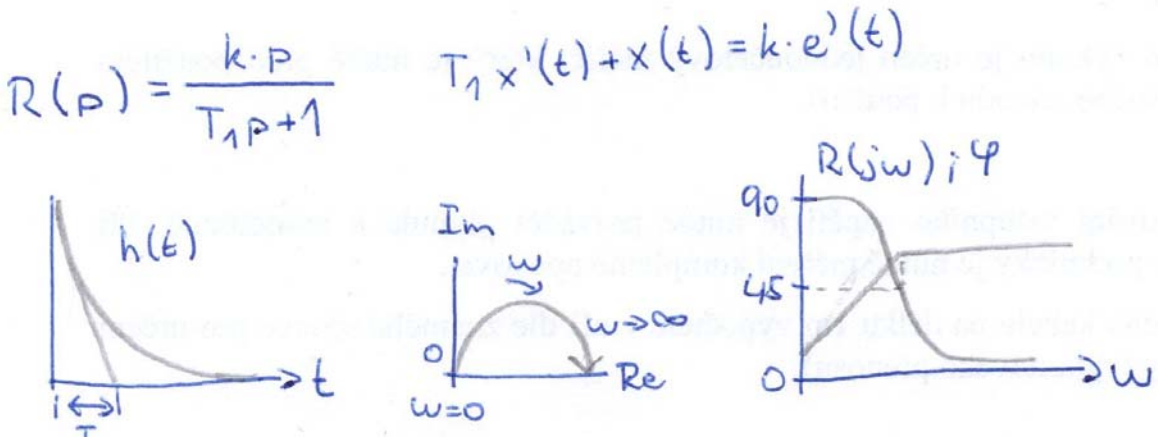
4 Derivační

4.1 Bez zpoždění – 0. řád

Fyzikálně nemožné



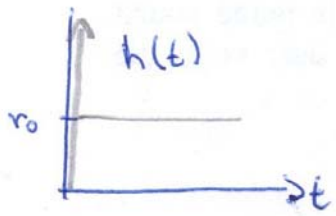
4.2 Se zpožděním 1. řád

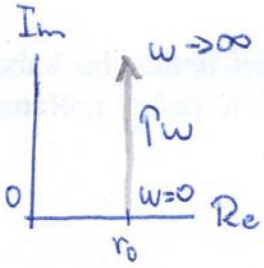


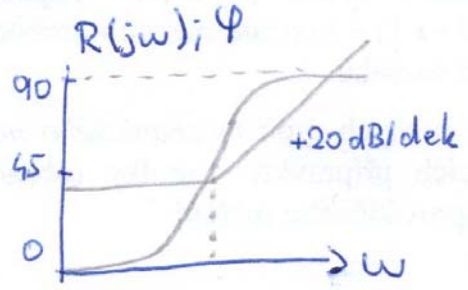
5 PD

5.1 Bez zpoždění – 0. řád

$R(p) = r_0 + k \cdot p$ $\quad x(t) = r_0 \cdot e(t) + k \cdot e'(t)$

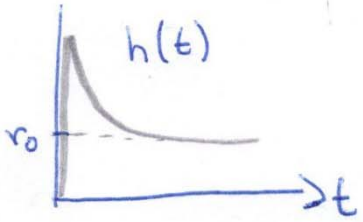


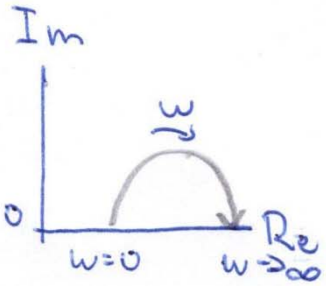


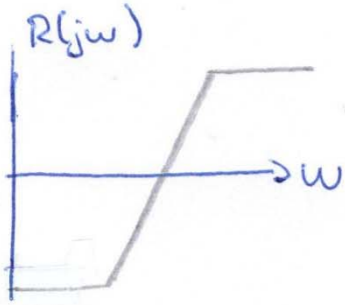


5.2 Se zpožděním 1. řád

$R(p) = \dfrac{r_0 + k \cdot p}{T_1 p + 1}$ $\quad T_1 \cdot x'(t) + x(t) = r_0 \cdot e(t) + k \cdot e'(t)$





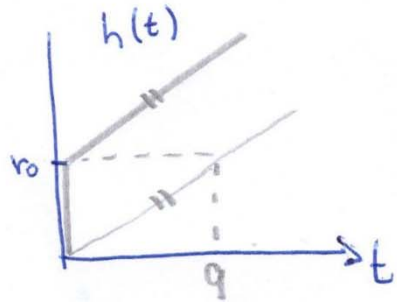


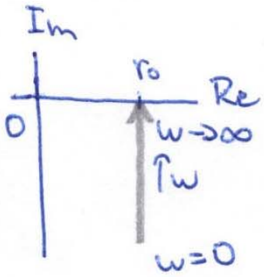
6 PI

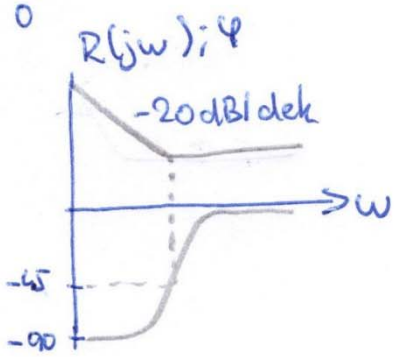
6.1 Bez zpoždění – 0. řád

Fyzikálně nemožné

$$R(p) = r_0 + \frac{q}{p} \qquad x(t) = r_0 e(t) + q \cdot \int_0^t e(\tau)\,d\tau$$

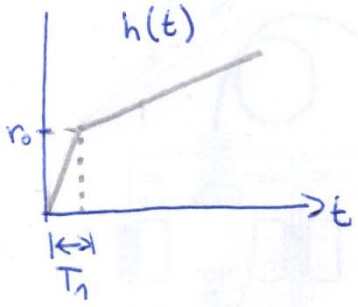


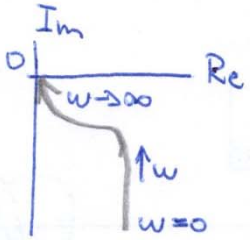


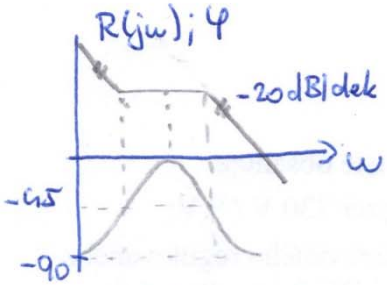


6.2 Se zpožděním 1. řád

$$R(p) = \frac{r_0 + \frac{q}{p}}{T_1 p + 1} \qquad T_1 x'(t) + x(t) = r_0 e(t) + q \cdot \int_0^t e(\tau)\,d\tau$$







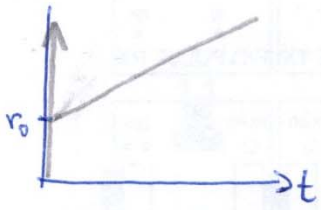
7 PID

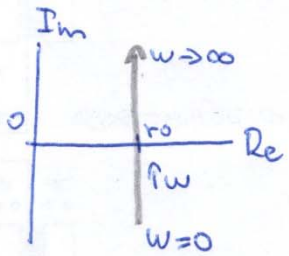
7.1 Bez zpoždění – 0. řád

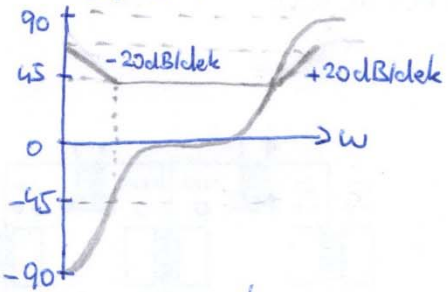
Fyzikálně nemožné

$$R(p) = r_0 + \frac{q}{p} + k \cdot p \qquad x(t) = r_0 e(t) + q \cdot \int_0^t e(\tau)\,d\tau + k \cdot e'(t)$$

$$R(p) = r_0 + r_1 \frac{1}{p} + r_1 \cdot p \quad \text{... paralelní tvar}$$

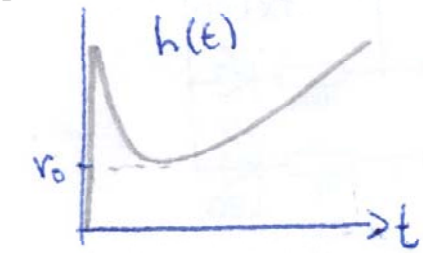


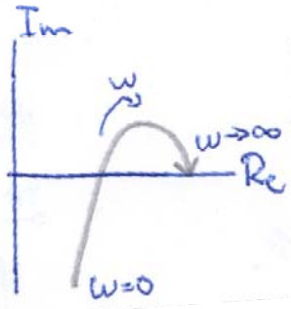


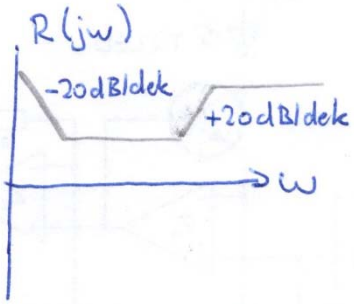


7.2 Se zpožděním 1. řád

$$R(p) = \frac{r_0 + \frac{q}{p} + k \cdot p}{T_1 p + 1} \qquad T_1 x'(t) + x(t) = r_0 \cdot e(t) + q \cdot \int_0^t e(\tau)\,d\tau + k \cdot e'(t)$$







7.3 Realizace s operačními zesilovači

